Denial Flow — Full Walkthrough, Paper Methods & Example

Part 1: Paper Quick Reference — What Each Paper Actually Does

Before diving into the flow, here's what each paper contributes in plain English:

Tag	Paper	What It Actually Does	What We Took From It
[Lop21]	López et al. 2021	Takes messy policy PDF text → parses sentences using dependency trees → extracts structured "benefit rules" (IF condition THEN coverage) into a knowledge graph	**The approach of converting policy text into structured IF-THEN rules.** Our Denial Intake and Policy Clause Verification agents use this idea to parse uploaded policy documents into queryable clause structures
[Pic22]	Picco et al. 2022	Trains a classifier with very few labeled examples to identify which sentences in a policy document describe a "benefit rule" vs. general text	**Few-shot classification of policy sentences.** Our Policy Clause Verification agent uses this concept — the LLM identifies which clause is relevant to the denial with minimal examples
[She24]	KAMEL 2024	Links free-text exclusion descriptions (e.g., "respiratory illnesses") to standardized ICD-10 codes using medical ontologies	**Medical entity linkage.** Our Case/Evidence Investigation agent uses this approach to connect the diagnosis extracted from clinical notes to the correct ICD-10 code and then check it against policy exclusions
[She24b]	Lui et al. 2024b	Extends KAMEL by adding LLMs + medical ontologies for end-to-end claims processing automation	**The hybrid ontology + LLM architecture.** We use this design pattern in Case/Evidence Investigation — the ontology provides structure, the LLM provides reasoning
[Pan24]	Pandey et al. 2024	Builds a multi-agent system where each agent handles one sub-task of Prior Authorization (extract checklist → check each item → give verdict with evidence)	**The multi-agent decomposition pattern.** This is literally our overall architecture — break the denial analysis into specialized agents, each handling one sub-task. Also: the "checklist with evidence" pattern feeds our Evidence Matrix

Tag	Paper	What It Actually Does	What We Took From It
[Ahs23]	Ahsan et al. 2023	Uses LLMs to search through patient clinical notes and retrieve specific evidence for a given medical query, with confidence scores	**Evidence retrieval from clinical text.** Our Case/Evidence Investigation agent uses this exact pattern — given a denial reason, search the discharge summary for supporting/contradicting evidence and return a confidence score
[Sbo24]	Sbodio et al. 2024	Builds a human-AI collaborative system where AI investigates claims compliance and a human reviewer validates findings before final decision	**The human-in-the-loop review pattern.** Our Human Review stage directly implements this — AI produces the analysis, human validates before the appeal is sent
[Kum10]	Kumar et al. 2010	Uses data mining (decision trees, rules) on historical claims to predict which claims will have processing errors	**Claims pattern analysis.** Our Financial/Admin Check agent uses this concept — rule-based checks on claim amounts, dates, and codes to catch errors/anomalies
[Liu26]	Liu et al. 2026	Creates InsLogicBench — formalizes insurance policy clauses into argumentation logic (claim → grounds → warrant → exception → rebuttal) using the Toulmin model	**Argumentation structure for appeals.** Our Applicability Reasoner and Appeal Agent use this Toulmin model: "The denial is wrong BECAUSE [grounds] SINCE [policy warrant] UNLESS [exception applies]"
[Pok26]	Pokharel et al. 2026	Converts policy text into symbolic logic rules (AND/OR/NOT) paired with a coverage-aware retriever, reducing LLM hallucinations by 44%	**Policy-to-logic conversion.** Our Applicability Reasoner directly uses this — convert clause conditions into logical rules (IF admitted AND policy_active AND NOT excluded THEN covered)
[Kis24]	Kiser et al. 2024	Uses GPT-4 to draft radiotherapy appeal letters, evaluates quality vs. physician-written letters	**LLM-based appeal drafting.** Our Appeal Agent uses this approach — the LLM generates the appeal letter grounded in policy clauses and evidence
[Mcc26]	DENIED-AI 2026	Pilot study using LLMs to reduce admin burden for denial management of imaging procedures	**Denial workflow automation pattern.** Our overall flow (intake → analysis → appeal) mirrors this study's workflow design
[Owo24]	Owolabi 2025	Trains an ML classifier to triage hospital admission denials — predicts which ones are worth appealing	**Denial triage/routing.** Our Denial Type Router uses this concept — classify the denial type and determine which investigation branches are needed

Part 2: Node-by-Node Flow Explanation

NODE 1: USER UPLOADS CASE

```
What happens: User uploads all documents related to a denied claim
Input docs:   Denial Letter + Policy + Claim Docs + Medical Docs +
              Bills + Pre-auth + Correspondence
```

This is just the entry point. No paper method here — raw document collection.

NODE 2: DENIAL INTAKE AGENT `[Lop21]`

```
What happens: OCR → Document Classification → Entity Extraction
Extracts:     Claim ID | Policy ID | Dates | Denial Reason | Cited Clause
CSV used:     claims_master.csv (to validate extracted entities)
              icd10_codes.csv (to validate any diagnosis codes found)
```

Paper method [Lop21]: López et al. showed how to take raw policy/claims text and extract structured entities using dependency parsing + an ontology. We use the same idea — the LLM reads the OCR'd denial letter and extracts structured fields (claim ID, dates, denial reason, cited clause number).

Concretely: The agent reads *"Your claim #CLM-2024-78432 dated 15/03/2024 for Total Knee Replacement has been denied under Clause 4.3 — Elective Procedures"* and extracts:

```
{
  "claim_id": "CLM-2024-78432",
  "denial_date": "2024-03-15",
  "procedure": "Total Knee Replacement",
  "denial_reason_raw": "Elective Procedures",
  "cited_clause": "4.3"
}
```

NODE 3a: DENIAL REASON NORMALIZATION `[NEW]`

```
What happens: Maps raw denial reason text → standard category
CSV used:     denial_reasons.csv
Output:       Standardized category (one of 7 types)
```

Takes the raw text *"Elective Procedures"* and maps it to the standard taxonomy:

Raw Text	→ Category	→ Subcategory
"Elective Procedures"	`MEDICAL_NECESSITY`	Elective/Non-Emergency

This is a **new contribution** not from any single paper — it's a lookup + fuzzy-match against `denial_reasons.csv`.

NODE 3b: CASE FACT RECONSTRUCTION `[Lop21] [Ahs23]`

```
What happens: Builds a complete fact profile of the case
CSV used:     claims_master.csv (patient/policy/financial data)
              medical_notes.csv (clinical narrative for this case)
Output:       Structured case profile
```

Paper method [Ahs23]: Ahsan et al. showed how LLMs can read clinical notes and extract relevant facts for a specific query. We use this to read the discharge summary and pull out medical facts.

Paper method [Lop21]: López's entity extraction approach structures the policy/claim facts.

Concretely builds:

```
Patient:   Ramesh Kumar, Age 58, Male
Policy:    ICICI Lombard Health Shield, Active 01/2023 - 01/2025
           Sum Insured: ₹10,00,000 | Room: ₹8,000/day
Claim:     ₹4,50,000 | Admitted: 10/03/2024 | Discharged: 16/03/2024
Hospital:  Apollo Hospitals, Hyderabad (Network: Cashless)
Diagnosis: Severe Osteoarthritis, Right Knee (ICD: M17.11)
Procedure: Total Knee Replacement (TKR)
Medical:   Conservative treatment failed over 18 months
           (physiotherapy, injections, pain medication)
           Bone-on-bone contact on X-ray
           Significant mobility impairment
```

NODE 4: DENIAL TYPE ROUTER `[Owo24] [Mcc26] + NEW`

```
What happens: Determines which investigation branches to activate
CSV used:     denial_reasons.csv (to understand denial type)
Output:       Which of 3 investigation paths to trigger
```

Paper method [Owo24]: Owolabi built an ML triage classifier that predicts appeal success based on denial type. We use this concept — based on the normalized denial category, the router decides which investigations are relevant.

Paper method [Mcc26]: DENIED-AI's workflow shows how different denial types need different analysis paths.

For our example (MEDICAL_NECESSITY denial):

```
Route decision:
  ✅ Policy Clause Verification  — need to check if Clause 4.3 applies
  ✅ Case/Evidence Investigation — need to find medical necessity evidence
  ✅ Financial/Admin Check       — always run as baseline
```

If it were a FINANCIAL denial (e.g., "exceeded sub-limit"), the router would skip the medical evidence branch and focus on financial analysis.

NODE 5a: POLICY CLAUSE VERIFICATION `[Lop21] [Pic22] [Pok26] [Liu26]`

```
What happens: Retrieves the cited clause from the policy and checks
              if the insurer applied it correctly
CSV used:     policy_clauses.csv (RAG vector search)
              icd10_codes.csv (check if diagnosis falls under clause)
Output:       Is the clause applicable? Correctly applied?
```

This is the **heaviest paper-method node** in the flow:

[Lop21] approach: The policy document has already been parsed into structured clauses in `policy_clauses.csv`. The agent queries: *"Retrieve Clause 4.3 from ICICI Lombard Health Shield"*

[Pic22] approach: The few-shot classifier identifies which sentences in the clause are the actual rule vs. just definitions/context.

[Pok26] approach: The clause is converted into symbolic logic:

```
Clause 4.3 — Elective Procedures:
  EXCLUDED IF:
    procedure IS elective
    AND NOT medically_necessary
    AND NOT emergency

  EXCEPTION:
    IF conservative_treatment_failed >= 6_months
    AND doctor_certifies_necessary
    THEN procedure is NOT elective → COVERED
```

[Liu26] approach: The Toulmin argumentation structure frames it:

```
CLAIM:     "TKR is excluded as elective"
GROUNDS:   "Clause 4.3 excludes elective procedures"
WARRANT:   "TKR can be scheduled in advance"
REBUTTAL:  "UNLESS conservative treatment has failed"
EXCEPTION: "If doctor certifies medical necessity"
```

Output for our example:

```
Clause 4.3 EXISTS in policy: ✅
Clause text mentions exception for failed conservative treatment: ✅
Insurer may have applied clause too broadly — did NOT check for exception
Status: POTENTIAL MISAPPLICATION
```

NODE 5b: CASE/EVIDENCE INVESTIGATION `[Pan24]` `[Ahs23]` `[She24]` `[She24b]`

```
What happens: Searches clinical documents for evidence that supports
              or contradicts the denial
CSV used:     medical_notes.csv (discharge summary text)
              icd10_codes.csv (validate diagnosis)
Output:       List of evidence items with status
```

[Pan24] approach: Pandey's multi-agent checklist pattern. The agent creates a checklist of what needs to be proven:

```
Checklist for "Medical Necessity" denial:
  ☐ Was conservative treatment attempted?
  ☐ For how long?
  ☐ Did conservative treatment fail?
  ☐ Is there imaging evidence of severity?
  ☐ Does doctor explicitly state necessity?
```

[Ahs23] approach: Ahsan's LLM evidence retrieval. For each checklist item, the LLM searches the discharge summary:

```
Query: "Was conservative treatment attempted?"
Found in discharge summary: "Patient underwent 18 months of
  physiotherapy, three cortisone injections, and daily NSAIDs
  with no significant improvement"
→ Status: FOUND | Confidence: 0.95
```

[She24] + [She24b] approach: KAMEL's entity linkage. The diagnosis "Severe Osteoarthritis" is linked to ICD-10 code M17.11 and checked against the policy exclusion list:

```
M17.11 → "Primary osteoarthritis, right knee"
Check against policy_clauses.csv exclusion list → NOT in permanent exclusions
Check waiting period → 24-month waiting period SATISFIED (policy active since Jan 2023)
```

NODE 5c: FINANCIAL/ADMIN CHECK `[Sbo24] [Kum10] [Joh21]`

```
What happens: Validates financial boundaries & admin compliance
CSV used:     claims_master.csv (amounts, limits, dates)
              nme_exclusions.csv (non-medical expense items)
              procedure_codes.csv (procedure code validation)
Output:       Financial discrepancies or admin issues
```

[Kum10] approach: Kumar's data-mining rules for claims error detection. Rule-based checks:

```
# Rule checks from [Kum10] pattern:
assert claim_amount <= sum_insured           # ₹4.5L ≤ ₹10L ✅
assert room_rent_per_day <= room_rent_limit  # ₹6,000 ≤ ₹8,000 ✅
assert admission_date >= policy_start_date   # 10/03/24 ≥ 01/01/23 ✅
assert length_of_stay is reasonable          # 6 days for TKR ✅
```

[Sbo24] approach: Sbodio's compliance investigation system. Scans the itemized bill against `nme_exclusions.csv`:

```
Bill Items:
  Surgery charges     ₹2,80,000  → PAYABLE
  Room (6 days)       ₹36,000    → PAYABLE (within limit)
  Implant (knee)      ₹1,20,000  → PAYABLE
  PPE Kit             ₹1,500     → NME EXCLUDED ⚠
  Registration Fee    ₹500       → NME EXCLUDED ⚠
  Dietary charges     ₹2,000     → NME EXCLUDED ⚠

  NME Deductions:     ₹4,000
  Net payable:        ₹4,46,000
```

Output: No financial grounds for denial. Only ₹4,000 in NME deductions.

NODE 6: DENIAL–CLAUSE–FACT–EVIDENCE GRAPH `[Liu26] [Lop21] [Sbo24] + NEW`

```
What happens: Assembles all findings into a structured graph
              Denial → Requirement → Case Fact → Evidence
```

[Liu26] approach: Uses the Toulmin argumentation model to structure the relationships:

```
DENIAL: "Excluded under Clause 4.3 — Elective Procedures"
  │
  ├── REQUIREMENT: "Procedure must be elective (not medically necessary)"
  │     │
  │     ├── CASE FACT: "Patient had 18 months failed conservative treatment"
  │     │     └── EVIDENCE: Discharge summary p.2, para 3 [CONTRADICTS denial]
  │     │
  │     ├── CASE FACT: "X-ray shows bone-on-bone contact"
  │     │     └── EVIDENCE: Radiology report [CONTRADICTS denial]
  │     │
  │     └── CASE FACT: "Doctor certifies TKR was medically necessary"
  │           └── EVIDENCE: Surgeon's notes p.1 [CONTRADICTS denial]
  │
  └── EXCEPTION: "Clause 4.3 has exception for failed conservative treatment"
        └── EVIDENCE: policy_clauses.csv, clause_id ICICI-EX-043 [SUPPORTS exception]
```

NODE 7: APPLICABILITY REASONER `[Liu26] [Pok26] [She24] [She24b]`

```
What happens: Applies logical rules to determine if denial holds
              AND/OR/NOT | Exceptions | Temporal Rules | Prerequisites
Output:       SUPPORTS denial / CONTRADICTS denial / INSUFFICIENT
```

[Pok26] approach: Pokharel's policy-to-logic conversion. Executes the symbolic rules:

```
# Logic from [Pok26]:
is_elective = procedure_can_be_scheduled AND NOT medically_necessary

# Check conditions:
conservative_failed = True   # 18 months evidence
doctor_certifies = True      # Surgeon's notes
imaging_severe = True        # Bone-on-bone X-ray

medically_necessary = conservative_failed AND (doctor_certifies OR imaging_severe)
# medically_necessary = True AND (True OR True) = True

is_elective = True AND NOT True = FALSE

# Therefore: Clause 4.3 exclusion does NOT apply
```

[Liu26] approach: InsLogicBench's defeasible reasoning:

```
Default rule:  "TKR is an elective procedure" → EXCLUDED
Exception:     "Unless conservative treatment failed" → NOT EXCLUDED
Conclusion:    Exception DEFEATS the default rule
Verdict:       CONTRADICTS DENIAL
```

NODE 8: EVIDENCE MATRIX `[Pan24] [Ahs23] [Liu26] + NEW`

```
What happens: Final structured summary of all evidence
```

[Pan24] approach: Pandey's checklist-with-evidence format:

Requirement	Fact	Evidence	Status	Source	Confidence
Must be elective	Failed 18mo conservative tx	Discharge summary p.2	CONTRADICTS	medical_notes	0.95
Must be elective	Bone-on-bone on imaging	Radiology report	CONTRADICTS	medical_notes	0.92
Must be elective	Doctor certifies necessity	Surgeon notes p.1	CONTRADICTS	medical_notes	0.97
Clause exception	Failed conservative tx > 6mo	Discharge summary	SUPPORTS exception	medical_notes	0.95
Financial validity	₹4.5L within ₹10L SI	claims_master	NO ISSUE	claims_master	1.00
Waiting period	Policy active 14 months	claims_master	SATISFIED	claims_master	1.00

NODE 9: DENIAL ASSESSMENT

```
What happens: Makes final verdict based on evidence matrix
Output:       One of 4 outcomes
```

Based on the evidence matrix:

- 3 items CONTRADICT the denial with high confidence (>0.90)
- 1 item SUPPORTS the exception clause
- No financial or admin issues

Verdict: `POSSIBLE GROUNDS FOR REVIEW` → The insurer appears to have misapplied Clause 4.3 by ignoring the exception for failed conservative treatment.

NODE 10: TARGETED RE-ANALYSIS `[NEW]`

```
What happens: Re-validates the specific grounds against all evidence
              to make sure the discrepancy is real, not a hallucination
```

The system double-checks:

- Does the policy PDF actually contain the exception language? → **YES** (re-reads `policy_clauses.csv`)
- Does the discharge summary actually mention 18 months? → **YES** (re-reads `medical_notes.csv`)
- Is the ICD code M17.11 actually NOT in the permanent exclusion list? → **CONFIRMED**

NODE 11: DEADLINE + ROUTE CHECK `[NEW]`

```
What happens: Checks appeal deadlines and determines handling route
```

```
denial_date = "2024-03-15"
today = "2024-04-20"
days_elapsed = 36
appeal_window = 30  # IRDAI internal appeal window (typically 30 days)

# For Indian insurance:
if days_elapsed <= 30:
    route = "INTERNAL_APPEAL"        # To insurer's grievance cell
elif days_elapsed <= 365:
    route = "OMBUDSMAN"              # To Insurance Ombudsman
else:
    route = "CONSUMER_COURT"         # To Consumer Disputes Forum

# Result: 36 days → OMBUDSMAN route (missed internal window)
```

NODE 12: APPEAL AGENT `[Kis24] [Mcc26]`

```
What happens: Drafts the appeal letter using LLM
CSV used:     policy_clauses.csv (cite exact clauses)
              medical_notes.csv (cite exact evidence)
              denial_reasons.csv (reference standard categories)
```

[Kis24] approach: Kiser showed that GPT-4 can draft clinically accurate appeal letters. We use the same approach — the LLM is prompted with all the structured evidence and generates a professional appeal.

[Mcc26] approach: DENIED-AI showed the workflow of LLM-assisted denial response. We follow the same pipeline.

Prompt structure (simplified):

```
You are drafting an appeal to the Insurance Ombudsman.

DENIAL: Claim CLM-2024-78432 denied under Clause 4.3 (Elective Procedures)
GROUNDS FOR APPEAL:
  1. Clause 4.3 has an exception for cases where conservative
     treatment has failed (see clause text: "...except where
     medical necessity is established through documented failure
     of conservative management...")
  2. Patient underwent 18 months of conservative treatment
     (Source: Discharge Summary, Page 2)
  3. X-ray evidence of bone-on-bone contact (Source: Radiology Report)
  4. Operating surgeon certifies medical necessity (Source: Surgeon Notes)

CITE the exact clause. CITE the exact evidence page. Be professional.
Address to: Insurance Ombudsman, Hyderabad.
```

NODE 13: GROUNDED APPEAL DRAFT `[Kis24] [Liu26]`

[Liu26] approach: The appeal follows the Toulmin argumentation structure:

To: The Insurance Ombudsman, Hyderabad
Re: Appeal against denial of Claim CLM-2024-78432

Dear Sir/Madam,

I am writing to appeal the denial of my health insurance claim by ICICI Lombard (Policy: HL-2023-XXXX) for Total Knee Replacement surgery performed on 10/03/2024 at Apollo Hospitals.

GROUNDS FOR APPEAL:

The insurer denied my claim citing Clause 4.3 ("Elective Procedures"). However, Clause 4.3 itself contains an exception (Sub-clause 4.3.2) which states: "Procedures shall not be deemed elective where the insured has undergone documented conservative treatment for a minimum period of six months without adequate clinical improvement."

SUPPORTING EVIDENCE:

1. My discharge summary (Page 2, Para 3) documents that I underwent 18 months of physiotherapy, three cortisone injections, and daily NSAID medication — all without significant improvement. This exceeds the 6-month threshold specified in Sub-clause 4.3.2.

2. Radiological imaging (attached) shows bone-on-bone contact in the right knee, confirming Grade IV osteoarthritis (ICD-10: M17.11), which is NOT listed under permanent exclusions in Section 5 of the policy.

3. My operating surgeon, Dr. [Name], has certified that the procedure was medically necessary (certificate attached).

CONCLUSION:
The insurer's application of Clause 4.3 is incorrect as it failed to consider the exception in Sub-clause 4.3.2. All conditions of the exception are satisfied. I request reversal of the denial and settlement of ₹4,46,000.

Attachments:
- Original Denial Letter
- Policy Document (Clause 4.3 highlighted)
- Discharge Summary
- Radiology Report
- Surgeon's Certificate of Medical Necessity
- Itemized Hospital Bill

Respectfully,
Ramesh Kumar

NODE 14: HUMAN REVIEW `[Pok26]` `[Sbo24]`

[Sbo24] approach: Sbodio's human-AI collaboration model. The AI has done its investigation — now the human reviews:

```
REVIEW CHECKLIST (shown in UI):
  □ Are the facts correct? (dates, amounts, names)
  □ Is the policy clause interpretation valid?
  □ Is the evidence accurately cited? (page numbers, quotes)
  □ Are the legal arguments sound?
  □ Is the appeal letter tone appropriate?
  □ APPROVE / REQUEST CHANGES
```

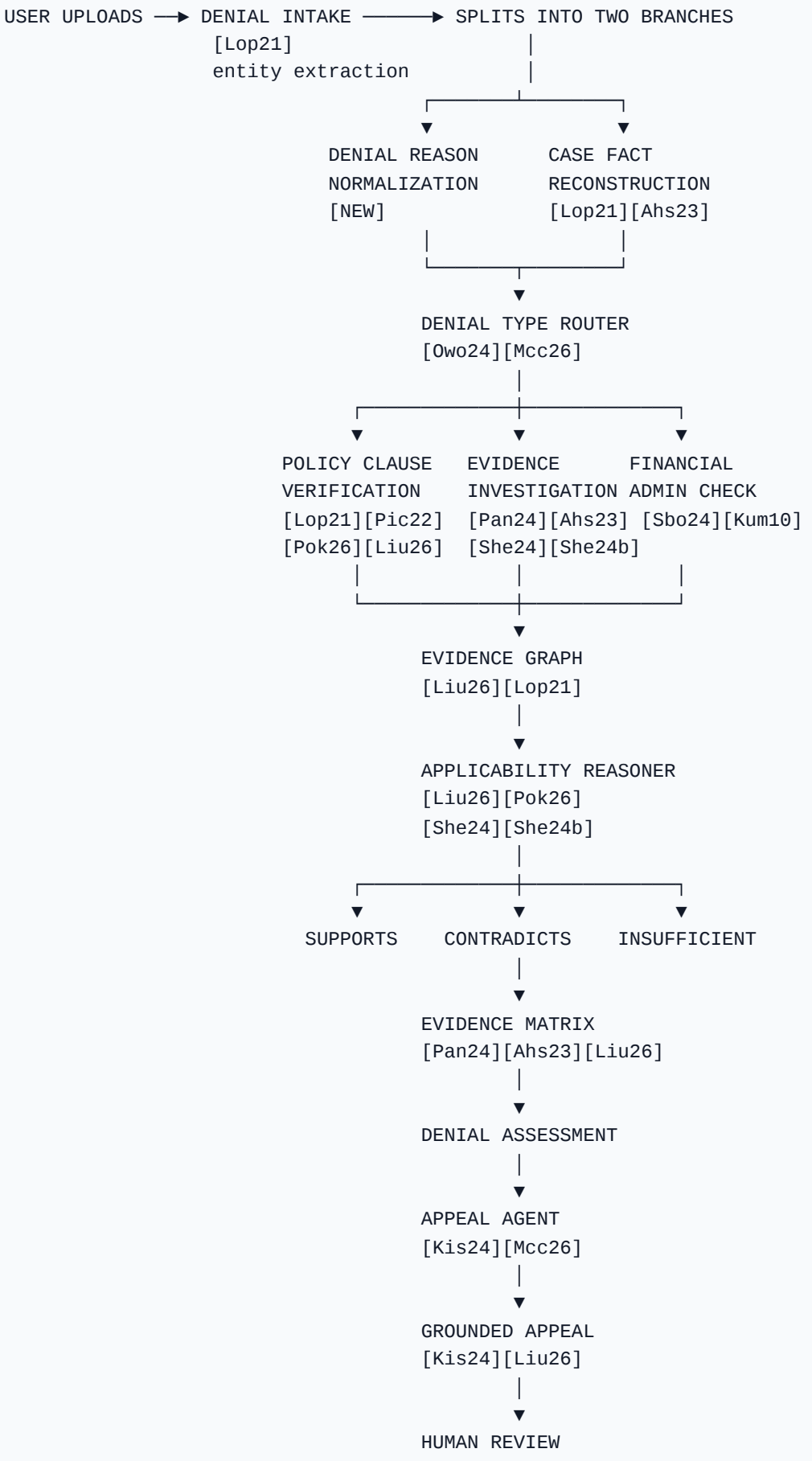
[Pok26] approach: The symbolic logic rules are shown alongside the appeal for the reviewer to audit — they can verify the logical reasoning step-by-step.

NODE 15: FINAL APPEAL + EVIDENCE PACK `[NEW]`

If human approves → system generates the final package:

```
📁 Appeal_CLM-2024-78432/
  ├── Appeal_Letter.pdf
  ├── Evidence_Pack/
  │   ├── Denial_Letter_Original.pdf
  │   ├── Policy_Document_Clause_4.3_Highlighted.pdf
  │   ├── Discharge_Summary.pdf
  │   ├── Radiology_Report.pdf
  │   ├── Surgeon_Certificate.pdf
  │   └── Itemized_Bill.pdf
  ├── Evidence_Matrix.csv
  └── Case_Summary.pdf
```

Part 3: Visual Summary — Paper → Node Mapping



```
                    [Pok26][Sbo24]
                          |
                          ▼
                  FINAL APPEAL PACK
                  [NEW]
```

Part 4: Example Summary Table

Here's the entire example condensed — one row per node:

Step	Node	Input	What Happened	Output
1	Upload	PDFs	User uploads denial letter, discharge summary, policy, bill	Raw documents
2	Intake	Raw docs	OCR → extracts claim_id, denial reason, clause 4.3	Structured entities
3a	Normalize	"Elective Procedures"	Maps to MEDICAL_NECESSITY category	Standard category
3b	Case Facts	claims_master + medical_notes	Builds patient profile: Ramesh, 58M, TKR, ₹4.5L	Case fact profile
4	Router	MEDICAL_NECESSITY	Routes to all 3 investigation branches	Route decision
5a	Policy Check	policy_clauses.csv	Finds Clause 4.3 HAS exception for failed conservative tx	POTENTIAL MISAPPLICATION
5b	Evidence	medical_notes.csv	Finds 18mo failed treatment, bone-on-bone, doctor cert	3 contradicting evidences
5c	Financial	claims_master + nme_exclusions	₹4.5L within ₹10L SI, ₹4K NME deductions	NO FINANCIAL ISSUE
6	Graph	All findings	Builds Denial → Requirement → Fact → Evidence tree	Structured graph
7	Reasoner	Logic rules	`is_elective = FALSE` because exception met	CONTRADICTS DENIAL
8	Matrix	All evidence	3 items contradict denial at >0.90 confidence	Evidence table

Step	Node	Input	What Happened	Output
9	Assessment	Matrix	Verdict: POSSIBLE GROUNDS FOR REVIEW	Recommendation
10	Re-analysis	Double-check	Confirms clause exception exists, evidence is real	CONFIRMED
11	Deadline	Dates	36 days elapsed → missed internal → OMBUDSMAN route	Route: Ombudsman
12	Appeal Agent	All evidence	GPT generates appeal letter citing clause 4.3.2, evidence	Draft letter
13	Draft	Letter	Toulmin-structured appeal with specific citations	Grounded appeal
14	Human Review	Draft	User checks facts, approves	APPROVED
15	Final Pack	All	Appeal letter + evidence pack + matrix + citations	📁 Ready to send